

Fig. 2: CPU power consumption during cold boot (Courtesy: anandtech.com)

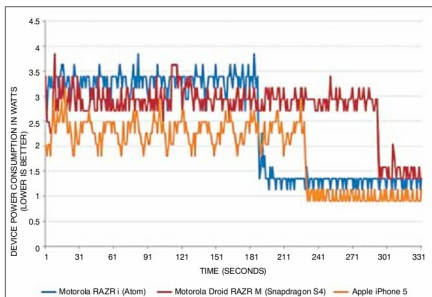


Fig. 3: Device power consumption vs time—Kraken benchmark 2 (Courtesy: anandtech.com)

consumed really low power while being powerful enough to handle WiFi, imaging and other simple stuff that we see on a smart phone or a tablet. This limited software options for them as x86-based applications would not run on these systems.

However, now that Intel has risen to the occasion and begun producing chips that offer comparable performance for even better battery life, the choice is now open to base your next product on either an x86-based processor or an ARM-based SoC.

In Fig. 2, the difference shown in average CPU power consumption

while doing a cold boot is significant enough to show improvement in x86. Tegra 3 pulls around 1.29W on an average compared to 0.48W by Atom.

"Atom also finishes the boot process quicker, which helps it get to sleep quicker and also contributes to improved power consumption," adds Shimpi in his analysis.

Obviously, Tegra 3 is not the only ARM SoC out there. NVIDIA has already released the latest Tegra 4 processor at CES 2013, which is based on 28nm process node compared to 40nm process node on its predecessor. The process node itself should lead

to a significant reduction in power consumption on the new SoC. Moreover, it uses an ARM Cortex A15-based 4-PLUS-1 quad-core CPU this time. It also features higher clock speeds, a 72-core GPU and 32-bit dual-channel LPDDR3 memory.

The chart in Fig. 3 shows how ARM-based A6 in iPhone 5 is a seriously competitive chip when it comes to power consumption—it is the lowest power consumer in the group. It is not sluggish in performance either, as you will see later on in this article.

Who performs better?

The answer is not straightforward because we are talking about two different things—complex instruction set computing (CISC) and reduced instruction set computing (RISC). Basically, the difference here is that CISC focuses on completing a function in just one instruction, while a RISC processor would use a couple of simple uniformly formatted instructions to do the same. At a glance, it might appear that RISC is less efficient. However, what you need to keep in mind is that the simplistic nature of an RISC instruction allows it to be processed in usually just one clock cycle.

A negative impact of RISC with all these instructions is that it uses far more RAM than CISC-based processors, which could be why CISC-based processors were all the rage until now. With the availability of inexpensive and fast RAM, this disadvantage has disappeared since the last half decade. Moreover, as mentioned earlier, the availability of 64-bit ARM solutions will enable the use of even more RAM, closing the gap further.

Even with the availability of competitive x86-based processors, most product vendors are still using ARM-based solutions to power their flagship devices. In performance, though, Windows 8 tablets based on Intel processors are more powerful than ARM-based Microsoft Surface tablets running Windows RT, for example.